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Well Performance with Operating Limits under Reservoir and Completion Uncertainties

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Abstract

This paper shows how to evaluate well performance under conditions of reservoir and completion uncertainty, while also considering the impact of completion decisions and operating constraints. This topic is important because it can help to reconcile the often-large gap between theoretical, short-term deliverability, and reliable, sustainable performance.

A table of operating constraints is presented, with examples regarding application of some of the constraints. This constraint table also includes consideration for the type of surveillance that is needed to apply the constraints. Discussion within the paper shows that the most appropriate types of operating constraints are often not being used. This paper also addresses appropriate operating limits for completions with sand control installed.

Completion selection and design influence operating constraints. Examples within the paper illustrate methods to determine appropriate operating constraints. The examples also demonstrate the potential advantage of frac-packed completions or long horizontal completions with regard to preventing fines migration and increasing longevity.

To aid with probabilistic performance forecasting, the paper includes a tabulation of published skin values, including permeability data when it was available.

Introduction

Financial risk analysis and petroleum reservoir performance under conditions of uncertainty have previously been addressed by many authors, beginning at least 40 years ago,^{1,2} and more recently in several analyses of the PUNQ-S3 case study.³ Other studies have addressed application of flexible production rules or constraints to simulation.⁴

Many operating constraints are needed to assure reliability and availability over the life of a project. A partial list of production constraints investigated in the literature includes erosional velocity,⁵ velocity through sand screens,⁶

critical drawdown pressure for completions without sand control,⁷ chemical composition and critical fluid velocity for fines migration,⁸ and minimum wellbore pressure and the impacts on scaling,⁹ casing collapse,^{10,11} screen collapse,⁶ or retrograde condensation.¹² The cited studies are not necessarily the earliest, and there have been many other studies of each of these constraints.

Both uncertainty and constraints should be considered to produce realistic forecasts, and examples showing how to address both are shown in this paper.

Transient conditions during startup and shutdown procedures are also very important considerations for operating a field in a reliable manner; however, constraints appropriate during startup and shutdown procedures are not addressed in this paper.

Well Performance under Uncertainty

Well performance is determined by reservoir and completion factors (uncertainties), and design choices and operating constraints (decisions). If reservoir performance forecasts don't adequately consider the consequences of both, significant errors can result.

Pseudo-steady state productivity, J , for a vertical well is given by the familiar relation:

$$J = \frac{.00708k_x h}{\mu B_o \left[\ln \left(\frac{0.472r_e}{r_w} \right) + \Sigma s \right]} \dots\dots\dots (1)$$

or, for a general well configuration in an anisotropic reservoir, by the following, where P_D is as defined in the paper:¹³

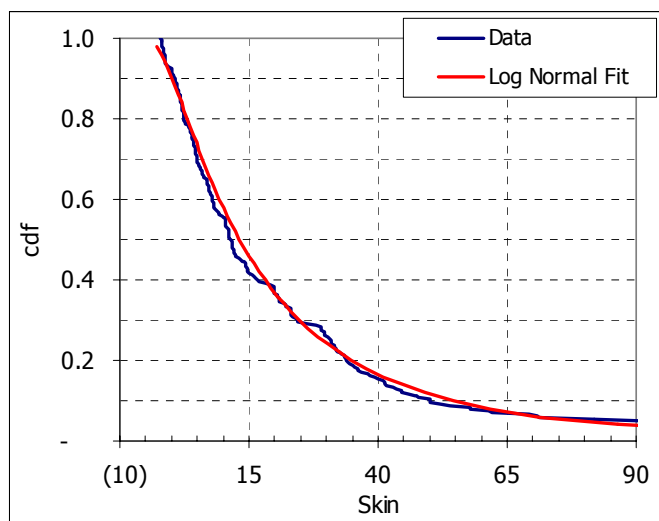
$$J = \frac{.00708\bar{k}L'}{\mu B_o \left[2\pi P_D \left(\frac{L'}{x_e} \right) + \Sigma s \right]} \dots\dots\dots (2)$$

In both equations above, some properties are uncertainties, other properties are the result of design choices, and skin is both. Properties such as formation volume factor, permeability, and viscosity are uncertainties. Although some of these uncertainty variables can be altered at times (i.e. viscosity can be lowered by supplying heat to the reservoir), the initial values of these parameters are usually intrinsic properties of the resource. Other values such as wellbore radius, drainage geometry, and completion length are the result of design choices. Skin is an uncertainty; however, the

range of skin varies substantially depending on the type of completion that is used. An extensive tabulation of skin values for vertical or deviated wells from published sources is shown in **Appendix A**. The data are believed to represent skin factors occurring before any remedial stimulation was applied; however, sufficient information was not always available to verify this. It is not known to what extent most of these data were corrected for other factors such as slant hole effect or partial penetration.

Completion types in **Appendix A** include Cased Hole Gravel Pack (CHGP), Cased Hole Frac Pack (CHFP), Cased Hole Perforated only (CHP), OH Gravel Pack (OHGP), and Cased Hole Fractured (CHF, screenless fractured wells). In cases where a paper is cited as a reference but only graphical data are found in the paper, the authors of the referenced papers were contacted to obtain the tabular data. In addition to the data in **Appendix A**, 349 other unpublished data points were available from proprietary sources to generate skin distributions for various completion types. Example skin distributions using the full 517 well dataset are shown for CHGP (**Figure 1**) and CHFP (**Figure 2**) completions. Another example shows the skin distribution using proprietary data for horizontal OHGP & OHSO completions with water-based drill-in fluid (**Figure 3**).

Figure 1 - CHGP (249 Wells)



The proprietary skin data for horizontal wells shown in Fig. 3 were calculated based on measured productivity indices and theoretical productivity indices for undamaged wells. There are several methods for calculating productivity of horizontal wells.^{13,14,15,16,17} For this work, the method of Economides' et al was used (**Eq. 2**), which considers permeability anisotropy and asymmetric well placement in the vertical plane, for example, wells placed very near the top of the reservoir. The table of shape factors published with this method has been extended to include low or high aspect ratio drainage shapes (aspect ratio > 4 or < 0.25) and short horizontal wells (dimensionless length < 0.25), as listed in **Table 1** and shown in **Figure 4** below.

Most of the horizontal well skins were based on data from a single field, producing 20° API oil. A net oil isopach map and vertical placement of the wells were used to assign

the various geometric factors, and a list of active completions at the time the PIs were measured was also used in determining the drainage areas. Care was taken to use only wells that were producing nil water at the time when the PIs were measured to avoid errors associated with multiphase flow effects.

Figure 2 - CHFP (184 Wells)

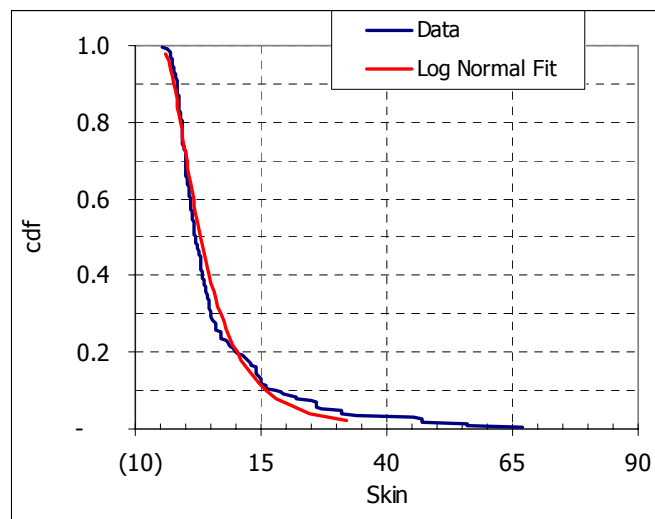
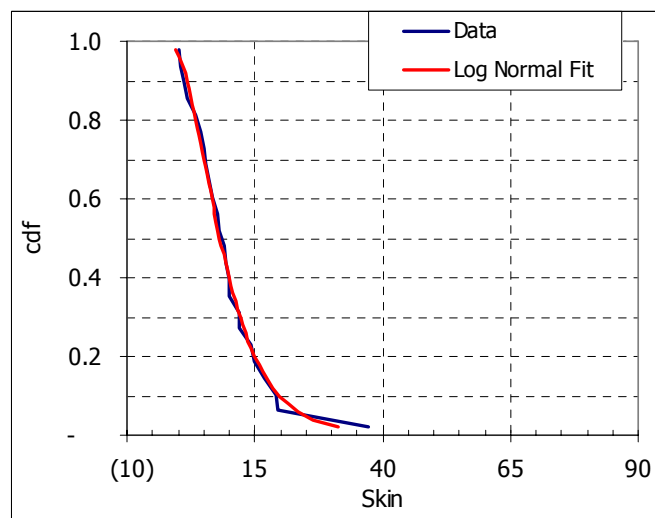


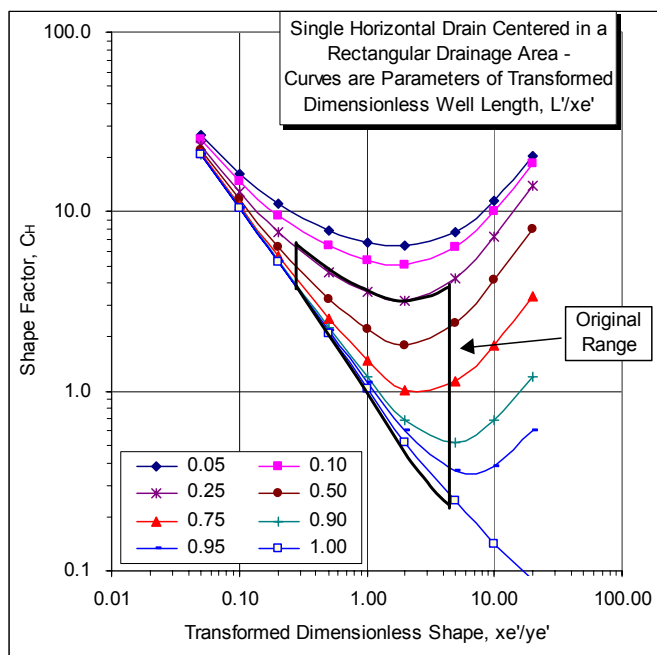
Figure 3 – Horizontal OHGP & OHSO w/ WB DIF (24 Wells)



PIs were calculated based on pressure drawdown measured at the heel rather than average pressure drawdown along the whole completion. This causes systematic error with the calculated PIs being lower than the actual PIs. **Eq. 2** shows skin is inversely proportional to PI, so skin would be proportionally lower based on the magnitude of this PI error. **Figure 5** shows production log data for a more recent horizontal well completion in the same field (field "X"). These production log data were obtained using two different down (actually "out") passes, using tools conveyed on 1-3/4" coiled tubing. For this well, the difference between the drawdown at the heel and the datum-corrected average drawdown was 20.4% for the high rate pass, and 10.2% for the low rate pass. The logged interval for this well was 1875 ft.

By contrast, the average wellbore length for the horizontal wells used for **Fig. 3** was 1133 ft, and the maximum length was 2062 ft. Only the PIs were available to the author of this report, and not the separate rate and drawdown data. This error is further complicated by the presence of the 1-3/4" coiled tubing during logging. The cross-sectional area of the coiled tubing was approximately 12% of the flow area inside the base pipe, so the pressure loss due to friction was slightly greater than it would have been without the CT present, and this error was proportional to the length of coiled tubing present across the completion.¹⁸ This error is smaller than the other error, but in the opposite direction. No attempt was made to apply a correction factor for these pressure losses in the wellbore. These pressure losses are believed to have resulted in an average error in the skins of less than 10%, for the field X well skin values.

Figure 4 - Shape Factors for *Economides et al* Method

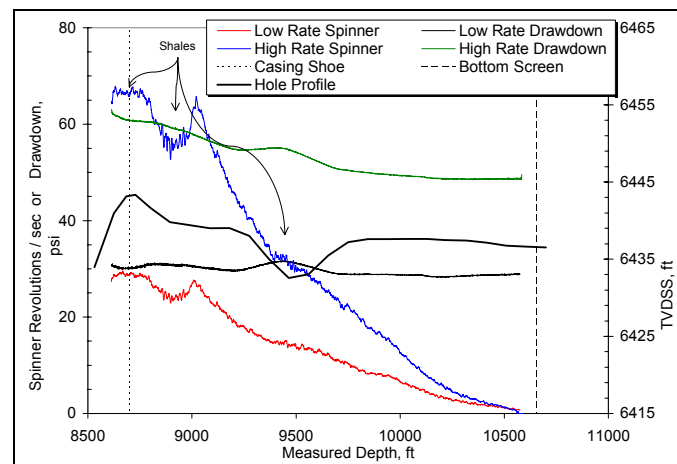


Although a comparison of Figure 2 and Figure 3 shows that CHFP completions have lower skins on average than OHGP or OHSO completions, the PI for a long horizontal OHGP or OHSO completion can be much higher than the productivity for a vertical well. A comparison of actual horizontal well PIs vs. theoretical PIs for undamaged vertical wells has been published.¹⁹

After uncertainties are identified, the next step in forecasting performance under conditions of uncertainty is to define the distribution for the uncertainty variables. Other authors have written about how to do this, including guidance regarding the best types of distributions to use for different types of data.^{20,21} The quality of probabilistic forecasts will be limited by the quality of the data and the underlying distributions used to characterize the independent uncertainties. Research has shown that humans consistently underestimate the range of uncertainty. Capen demonstrated that even highly technical people typically underestimate the range of uncertainty, even after being told that their ranges are usually too small before being asked to provide their

estimates.²² Another important point is that some of the uncertainty variables in the productivity equation may be dependent on other variables that are not in the equation. For example, the ability to correctly assess the probabilistic distribution for in situ viscosity depends not only on fluid sampling procedures and measurement techniques, but also on correctly assessing the range of reservoir temperature.

Figure 5 - CT-Conveyed Production Log Data



After distributions are defined for all the uncertainties such as permeability, viscosity, and skin, probabilistic calculations can be done. The next steps in the calculation process will depend on whether completion design or reservoir-level modeling is being done.

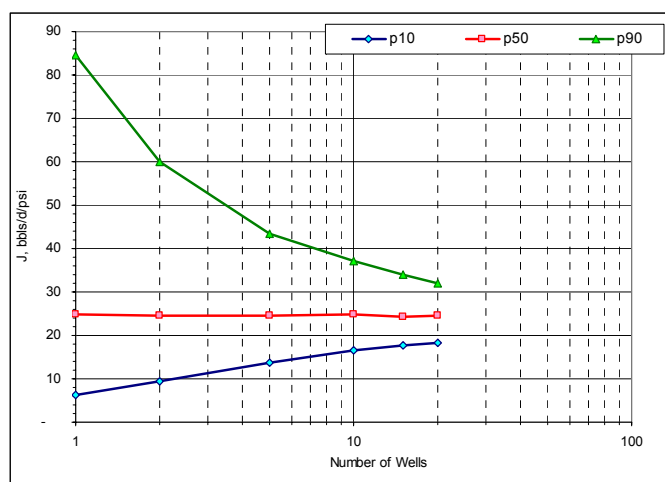
For completion design purposes, Monte Carlo calculation methods can be used with **Eq. 1** or **Eq. 2** to establish the probabilistic distribution for productivity. The productivity index can be used along with appropriate constraints (see the next section) to determine the operating condition ranges for the wells.

For reservoir simulation, the variables described above, plus additional uncertainty variables such as porosity, water saturation, relative permeability, etc. can be used with experimental design techniques to quantify the most important variables. A polynomial proxy can then be defined to predict reservoir performance over a broad range of conditions using Monte Carlo simulation. This method will help to identify what are the most critical variables controlling reservoir performance, and help to focus further evaluation efforts. When doing this modeling, it is possible to use productivity or skin uncertainty for individual wells; however, it may be impractical to designate a separate uncertainty variable for well performance (skin, completion efficiency, or productivity / injectivity) for every well in the model. If that is the case, then the composite performance uncertainty for all the wells or groups of wells must be used. When this is done, the range of uncertainty for average well performance will narrow as a function of the number of wells being considered in the well group, as shown in **Figure 6**.

It may be necessary to also account for the completion strategy and decisions that are made during the completion process when calculating the probabilistic productivity. It may not always be desirable to test wells before moving the rig off of a well. For example, for completion methods that have a low probability of requiring

remedial action after the initial installation (e.g. CHFP), the preferred strategy may be to complete each well without production testing the well until after the rig has moved to another well. For non-fractured completion methods when the cost of testing wells is very high and the risk of moving ahead without production testing is unacceptable, it may be preferable to pump a matrix stimulation treatment on every well, sans testing. A decision analysis economic model may be needed to establish the best strategy to follow (see example, **Figure 7**). The outcome of the decision analysis could alter the expected distribution for skin or productivity. For example, the decision analysis may show that testing each horizontal OHGP well before the rig leaves is the preferred option, and wells with skin greater than +10 will be acidized, but only if J is less than the target productivity of 25 bbls/d/psi. Of course an acid job can fail, and probability of success for the treatment should also be considered; however, it is obvious that the final distribution for skin would be altered by such a strategy. After a decision is taken regarding what strategy will be followed, this may alter the expected final distribution for well productivity.

Figure 6 - Example of Multi-Well Horizontal OHGP J Distribution



Operating Constraints

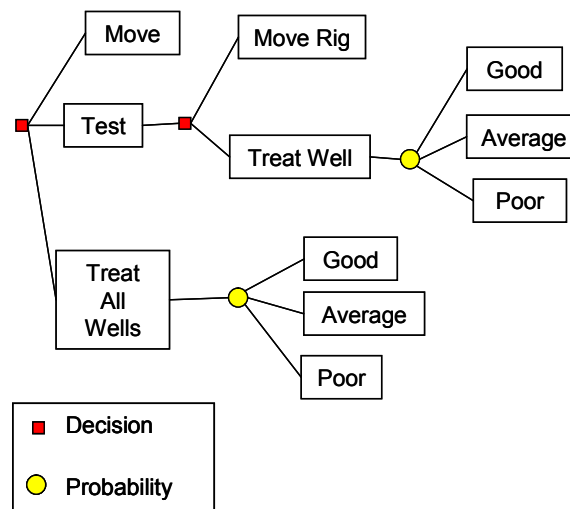
Many different constraints can impact the design and operation of oil and gas projects. Some of the constraints that may be applied to various aspects of the project are listed in **Table 4**, along with the issues addressed by the constraints, and the regions where the issues may occur. Constraints that are used to provide increased reliability, but could also have an impact on limiting production are the focus of this section. These include bottom-hole pressure based limits such as maximum drawdown across a completion, and velocity-based limits such as screen flux or fluid velocity in the tubing. In addition, a limit based on drag forces in the formation is proposed.

Total Drawdown

For a production well, total drawdown is the pressure drop from a no-flow or constant pressure boundary to the interior of the wellbore, on the downstream side of completion equipment such as gravel pack screens or sliding sleeves. Mathematically, it is the pressure at the drainage radius minus the flowing bottomhole pressure. Total drawdown can be

used as a constraint to help prevent two serious completion problems: collapse, and screen erosion. Prudent control of total drawdown may also help avoid problems with fines migration.

Figure 7 - Stimulation Decision Tree



Total drawdown can be valuable for preventing collapse of casing or sand screens, because total drawdown is the maximum pressure differential that could be applied across a section of the completion where no inflow is occurring. The most extreme example of this would be a cased-hole completion juxtaposed against a shale section or other very low permeability section where virgin pressure is maintained as the adjacent reservoir depletes. Studies have shown that pressure drop across screens is normally very low; however, if sand screens become almost completely plugged so that little or no flow can occur, the screen may be vulnerable to collapse. Failure will usually occur by deformation of the sand control media, resulting in sand production, rather than by collapse of the base pipe.⁶

Total drawdown can also serve as a constraint to guard against screen erosion. Screen erosion is a localized problem that can occur if the flowrate is sustained at an excessive level at any point along the screen. So-called “hot spots” are just small regions with relatively high inflow rates, perhaps caused by some combination of higher than average permeability and lower than average skin damage. One strategy to handle this problem would be to assume that some part of the completion is completely undamaged, or perhaps even stimulated in the case of a fractured or highly deviated completion. Under this assumed low-damage or stimulated condition, the maximum total drawdown that can be applied without exceeding the erosional velocity inflow limit of the screen can be determined. Very little has been published in the public domain regarding erosional limits of specific sand control devices, although some proprietary, collaboratively funded studies have been done.²³

An additional complicating factor is that the method described above will not always give the highest possible flowrate that could occur in the well. A mostly damaged completion with some relatively undamaged areas is analogous to a partially completed well as studied by Goode

and Wilkinson.¹⁷ In one of their examples (their Figure 4, Case 1), they found a well that is only completed over 20% of its length would have 74% of the productivity of a fully completed well. This implies that the average inflow velocity in the partially completed well would be 3.7 times greater than the average inflow velocity in the fully completed well. Production log data are needed on a case-by-case basis to fully assess the impact of this converging flow phenomenon on assignment of operating constraints. In the absence of production log data, assumptions need to be made regarding the amount and distribution of open interval, so that the effect of flow convergence can be considered when calculating the expected velocity.

By the same logic described above, total drawdown can also be used as a leading indicator to avoid fines migration. Additional discussion follows in the sections on Velocity and Viscous Drag.

Completion Drawdown

Completion drawdown is the actual total drawdown minus the ideal drawdown that would occur for an undamaged well. This extra drawdown (which may also be negative in the case of a stimulated well) is often a composite of many different effects, which have been thoroughly discussed by many authors cited in a distinguished author series article.²⁴ Also, other authors have discussed whether skin, completion drawdown, or some other measures should be used to benchmark completion quality.²⁵

Completion drawdown can be used to gauge the quality of a new completion, and it can also provide valuable information about changes in performance over time, such as gradual plugging of sand screens with fines, or near-wellbore precipitation. Skin or completion drawdown is also needed to forecast performance of a well. Completion drawdown appears to be valuable as a lagging indicator and a diagnostic tool; however, it may have lesser value than total drawdown as a leading indicator to prevent the onset of completion problems. Completion drawdown is controlled by average productivity, average skin, and outflow performance. As previously described, collapse is controlled by the most-damaged or non-contributing intervals, and screen erosion and fines migration are controlled by the least damaged intervals.

Formation Drawdown

Formation drawdown is simply the ideal drawdown that would occur if no skin damage were present. Mathematically, it is equal to total drawdown minus completion drawdown. As with total drawdown, average formation drawdown is useful to predict velocities, and as such, it is a leading indicator for problems such as erosion and fines migration. Increasing formation drawdown may indicate problems such as relative permeability effects or compaction-related permeability decline.

Bottomhole Pressure

For reservoirs with strong water drive or full voidage replacement injection, bottomhole pressure may be a satisfactory substitute for total drawdown or formation drawdown, to be used for prevention of screen erosion or fines migration.

For many inorganic scales, flowing bottomhole pressure in the wellbore is directly related to the scaling potential. Bottomhole pressure also controls the effective stress borne by the formation, which will have an impact on formation collapse and sand production. Many studies have shown that for wells with no sand exclusion device preventing sand ingress, a critical pressure can be calculated below which sand failure and sand production are expected to occur. Permeability can decline as a result of compaction; therefore, bottomhole pressure may also have a large impact on the formation permeability near the wellbore. For non-fractured completions, permeability reduction near the wellbore can severely impact productivity.

When non-producing intervals are present within the completed interval, bottomhole pressure can also serve as a constraint that will help to prevent collapse of the completion or the casing.

Velocity

Average velocity (flux) is an important constraint for several different production issues. Erosion will occur if velocity exceeds a critical value, and in multiphase flow, unstable flow can occur if velocity is too low. For projects that will produce under a declining reservoir pressure, trade-offs may be required between high initial flow rates and longer-term flow stability. Just as erosion will remove metal from pipes and fittings, it will also remove metal from the edges of the openings in sand screens. When the openings have been enlarged sufficiently that they will no longer hold back the gravel, failure of the gravel pack will occur.⁶

Field experience has shown that flow through some formations that exceeds a critical flow velocity will cause the movement of colloidal clay particles ("fines"). This has been confirmed by laboratory testing.⁸

Viscous Drag

Researchers have studied the interaction between forces acting to restrain or cause movement of a monolayer of ideal spherical particles on smooth surfaces.²⁶ In that ideal case, the mechanism of movement was found to be by rolling, rather than by sliding or lifting.

It is obvious that fines will also migrate through a reservoir when forces acting to move the particles exceed the restraining forces acting to hold them in place. Exact solution of this problem is very difficult because the flow geometry and the particle interactions are very complex; however, some insight into the process may be gained by looking at a simpler case that may capture the key hydrodynamic issues. The simple case of drag force imparted to a free-floating spherical particle of assumed diameter can be determined as follows²⁷:

$$F_d = \left(\frac{\pi}{8} \right) \frac{\rho C_d}{32.174} \left(\frac{D_p v}{12} \right)^2 \dots\dots\dots (3)$$

The drag coefficient is obtained from the following:

$$C_d = \begin{cases} 24 / \text{Re} & \text{Re} \leq 2 \\ 18.5 / \text{Re}^{0.6}, & 2 < \text{Re} \leq 500 \text{ (4)} \\ 0.44, & 500 < \text{Re} \end{cases}$$

For this calculation, Reynold's number is:

$$\text{Re} = \frac{124\rho v D_p}{\mu} \text{ (5)}$$

For flow through a porous medium, superficial velocity for a given mobile phase is:

$$v = \frac{0.000065Q}{a\phi S} \text{ (6)}$$

Traditionally, fines migration tendencies have been evaluated in the laboratory using critical velocity tests.⁸ These tests are done with reservoir fluids and at reservoir temperatures, to try to match the hydrodynamic conditions in the formation. An alternative approach could be used. The test could be conducted with any fluid of known viscosity. After the test is conducted, if fines movement was verified during the test, the critical drag force on an assumed spherical particle of appropriate size could be determined, and field rates that will not exceed this drag force could also be calculated. The particle diameter could be based on an analysis of the formation particle size distribution, or some arbitrary standard size could be used for the calculation. Examination of **Eq. 3 – Eq. 6** shows that at very low Reynold's number, the drag force on a free floating particle is directly proportional to the product of velocity and viscosity. In other words, they have equal effect.

These concepts are also illustrated in **Table 2**. Results A and B are actual laboratory core flood results for two different formations, with calculated results for drag force. Note that no critical velocity was observed during the test for well A. Hypothetical results B1 and B2 show the expected change in theoretical results for heavier oil and lighter oil respectively, based on drag for a free floating particle.

Although superficial velocity and drag force may be the best indicators for screen erosion and fines migration respectively, calculating the distribution for these requires production logging, or perhaps at some future time, distributed rate sensing. When frequent pressure transient data are available and only sparse or non-existent production profile data, it is more feasible to make some assumptions about the flow profile and calculate the maximum allowable drawdown based on completion collapse, casing collapse, screen erosion, and fines migration. The maximum drawdown is then just the minimum value that will cause any one of these problems to occur. It is also possible to make assumptions about the flow distribution and the skin to determine the maximum velocity that could occur in a given field, and then use that information to determine the range of flowrates needed for laboratory testing to reproduce the drag forces that may occur in the field.

Numerous studies including some recent investigations^{28,29} have sought to explain observed changes in sanding tendencies following water breakthrough. Segregated flow through pores and drag forces may help explain why problems associated with fines migration seem to also occur immediately after water breakthrough. Clay fines are often present as pore lining materials. For water-wet formations, fine clay particles may be present predominantly or even exclusively within the water saturated area lining the pores. If this is true, it may help to explain why fines movement is often not observed until after water breakthrough: there simply may be no fluid movement within the region where the fines exist. It may also mean that critical drag force should be determined in the laboratory for two different mobile fluids. One test should be conducted with hydrocarbon flowing through a core at connate water saturation. A second test should be conducted with water flowing through a core at residual oil saturation. Note, to use this simple approach, more than one phase may be present; however, only one phase should be mobile, and the saturation of the mobile phase should always be used to calculate the velocity in **Eq. 6**.

There may be an added advantage to using drag force to characterize fines migration tendencies. Clearly fines movement is a function of many complex factors, including hydrodynamics, particle size and shape, restraining forces caused by cementing materials or other factors, etc. By calculating a drag force based on an arbitrary (and perhaps standardized) particle size, some of the variability in fines migration behavior due to hydrodynamic factors may be normalized out. In other words, if the rock is similar and the drag forces are similar, then similar fines migration behavior may be expected regardless of whether the fluids have the same viscosity. Stated another way, the industry may be able to learn what range of drag forces can be expected to result in problems for a given formation type. This may provide a starting point for determining when fines migration may be a problem, and when further testing may be needed.

Well Performance with Uncertainty and Constraints

As discussed above, average productivity for a well can be calculated under conditions of uncertainty by evaluating the distribution of the uncertainty variables, and using Monte Carlo techniques to evaluate the probabilistic distribution of the productivity. As also discussed above, appropriate operating limits to address velocity-related issues should be based on the least damaged intervals, and they may also need to address converging flow for partially completed or partially damaged wells.

This section of the paper discusses example productivity and operating constraint calculations to illustrate the concepts previously discussed, and also how the choice of completion type will affect the appropriate range of operating conditions. Nine different example cases are considered. All cases are for the same reservoir geometry and rock properties. Five cases are for the same light oil reservoir fluid properties, and four cases are for heavier oil. For all these cases, the calculations are based only on liquid velocity; however, obviously gas flow velocity can play an important role below the bubble point. Flowing bottomhole pressures below the bubble point were corrected using Vogel's method; however,

no allowance was made for free gas flow in the formation. The skin distributions used are based on the full 517 well dataset, including **Appendix A** and proprietary data. Uncertainty distributions are shown in **Table 3**. Other data and results are summarized in **Table 5**.

Only three constraints were used in these examples: casing or completion collapse limits, screen erosion, and fines migration. Other constraints such as tubing erosion would also need to be considered for a real case. Also, it is assumed that the completion equipment and artificial lift system (if necessary) can achieve the maximum desirable total drawdown.

These example data do not represent any field in particular; however, they may be representative of low-strength reservoir rock with relatively shallow burial depth, underlying moderately deep water, and in a normally pressured basin. A key assumption for all these calculations is the minimum expected localized skin in any part of the completed interval. It is assumed that this “least damaged” zone will control the occurrence of fines migration, which could eventually lead to screen failure or poor performance. Another important assumption is the critical drag force. This is just an example obtained by assuming a value lower than the two examples shown in **Table 2**. Also, it was assumed for these examples that no stimulation will be done following the completion, so the skin distributions used will be those found in **Appendix A**. The distribution used for the vertical OHGP well was the same as the distribution derived from the 24 water-based horizontal OHGPs drilled with water-based fluid from the proprietary database. Were a sufficiently large dataset available, it is believed that the skin distribution for vertical or deviated, relatively short OHGPs would be lower than the skin distribution for horizontal OHGPs found in this report.

Case A – Vertical Well, Light Oil, Cased-Hole Perforated (CHP) Completion. For this as well as most of the other examples, the minimum P_{wf} constraint is controlled by fines migration. For this example, the surface area of the perforation tunnels was compared to a hypothetical open hole at a radius equal to the perforation tips. The perforation tunnel surface area was lower, so that was used for the formation flux calculation. The available drawdown for this completion is low, because the flow area between the perforation tunnels and the formation is low.

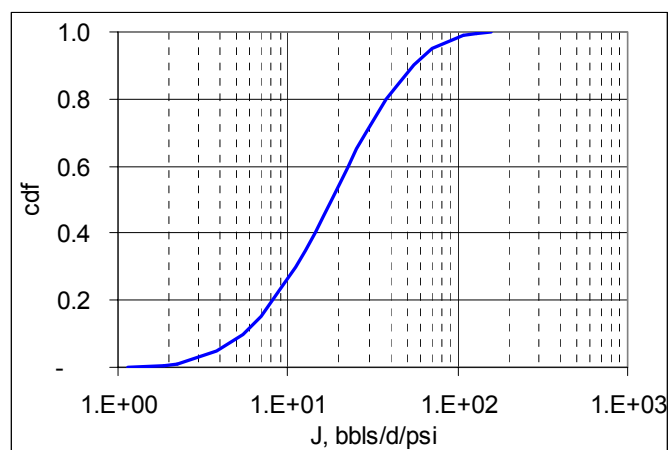
Case B – Vertical Well, Light Oil, Cased-Hole Gravel Pack (CHGP). For this example, the maximum formation flux was calculated based on the minimum of two areas: the sum of hemispherical areas representing the perforation cavities, and the area of a hypothetical enlarged wellbore at the tip of the perforation cavities. The hypothetical wellbore area was used because it was smaller. Although gravel contamination in the perforation tunnels and other issues yield a higher skin for this completion type than for the CHP completion, the allowable drawdown is significantly greater because of the larger minimum formation flow area. This results in significantly increased deliverability without exceeding the recommended minimum P_{wf} .

Case C – Vertical Well, Light Oil, Open-Hole Gravel Pack (OHGP). This example is for a completion with, on average, a lower skin distribution than a CHGP;

however, the minimum formation flow area and consequently the allowable drawdown and maximum production are significantly lower.

Case D – Vertical Well, Light Oil, Cased-Hole Frac Pack (CHFP). This completion has the highest maximum production of all the cases considered. It has a very high productivity, and also a high allowable drawdown. The minimum P_{wf} is limited by screen erosion rather than by fines migration. This is primarily because of the high area between the fracture and the formation. This may help to explain apparent greater longevity for CHFP completions as compared to most gravel packs.

Figure 8 - Productivity Distribution for Example CHGP



Case E – Vertical Well, Heavy Oil, Cased Hole Frac Pack (CHFP). The vertical, heavy oil well with a frac pack has excellent capacity compared to the other heavy oil cases (second only to the 5000' OHGP, case H); however, very high drawdown is needed to achieve these high rates. The P_{wf} is limited by screen collapse capacity in this case. Note that the screen collapse rating shown is just an example, and there is little doubt that screens stronger than this could be obtained. This design might also require additional operating energy in the form of artificial lift to obtain the high drawdown, and the completion could have additional vulnerability to scale or other deposits because of the low P_{wf} .

Case F – Short Horizontal Well, Light Oil, OHGP. This example case has the highest productivity of all the cases; however, the maximum production rate is less than the light oil frac pack case because the minimum P_{wf} is higher for this case. P_{wf} is again limited by fines migration. This is exacerbated by converging flow. A convergence factor of 4 was arbitrarily assigned to this case to account for high flux rates near the heel and the toe for this horizontal well that is relatively short in relation to the drainage area. Additional study would be needed to determine the best convergence factor to use for this case. Examples demonstrating high flux near the extremities of a horizontal have been published.³⁰

Case G – Short Horizontal Well, Heavy Oil, OHGP. A short horizontal well yields low productivity and low drawdown capacity for this case. The completion is also limited by fines migration. A convergence factor of 4 was also used for this case.

Case H – Long Horizontal Well, Heavy Oil, OHGP. This case has the highest rate capacity for all the

heavy oil cases. Productivity is comparable to the vertical, light oil case with an OHGP (case C). This completion is limited by fines migration. This completion contacts more of the reservoir sand than the heavy oil CHFP (case E); therefore, some differences in sweep efficiency may occur for a waterflood or strong water drive.

Case I – Long Horizontal Well, Heavy Oil, Partially Damaged OHGP. This case again uses the example from Goode and Wilkinson, Figure 4, Case 1.¹⁷ For this case, the well is partially limited by reduced PI, but it is more severely limited by the reduced available drawdown that results from flow convergence (“hotspots”). This example illustrates the importance of keeping the screens clean during completion. While no examples are shown in this paper, it seems apparent that partial penetration or “hotspots” could be quite undesirable and exacerbate fines migration problems in vertical completions also. A convergence factor of 3.7 was used based strictly on 74% PI divided by 20% flow area as compared with Case H. It would be easy to see why an even higher factor could be necessary. Further modeling could reveal that each short producing section could experience relatively higher inflow near the ends and less in the middle.

Conclusions

1. Well and reservoir performance forecasts must consider reservoir and completion uncertainties, completion decisions, and appropriate operating constraints that will assure sustainable deliverability over the life of the asset.
2. Critical drag force may be a better property than critical velocity for characterizing the onset of fines migration in laboratory experiments; however, additional study is needed to develop the use of this method. If this critical drag force or some other function incorporating viscous forces proves to be a useful method, it may not be necessary to use field fluid samples or field temperature conditions to determine critical drag force. Further study may show that once the critical drag force has been obtained, critical velocity under reservoir conditions can be calculated.
3. Distributed flow rate data is useful to derive operating constraints to avoid screen erosion, and also to avoid fines migration if the critical velocity or critical drag force have been measured.
4. Total drawdown or minimum bottomhole pressure are useful constraints to help prevent casing or screen collapse.
5. Total drawdown, formation drawdown, or bottomhole pressure are useful as constraints to avoid screen erosion and fines migration. Converging inflow caused by partially completed intervals or partially damaged wells may have a large impact on velocities, and this must be considered when estimating flow profiles in the absence of log data. Screen erosion and fines migration damage types may be most likely to occur in areas of highest velocity, which may originally be the least damaged areas of the well.
6. Completion drawdown is more useful as a lagging indicator that may be helpful for diagnosing problems that have already begun to occur.

7. Segregated flow and the distribution of fines within a pore structure may help to explain why some formations exhibit severe changes in productivity soon after water breakthrough. After water breakthrough, water begins moving in the region where the fines occur, and will mobilize the fines if the critical drag force is exceeded.
8. Cased hole frac packs may provide advantages in avoiding fines migration relative to other completion types because of the large contact area between the fracture face and the formation.
9. Long horizontal completions can provide high total rates comparable to vertical well frac packs at a lower drawdown by providing higher productivity. The long contact interval with the reservoir may yield improved sweep efficiency for some projects with unfavorable mobility ratios, and the low drawdown may help avoid precipitation problems or other problems associated with low bottomhole pressure.
10. By using concepts in this paper to analyze expected performance under conditions of uncertainty and set appropriate operational constraints, project personnel can reach a common understanding of well performance expectations and rationale for the operating limits.
11. An extensive table of previously published skin values for different completion types that can be used to assist with probabilistic evaluations has been included in this paper.
12. The shape factor table for a single horizontal well centered in a rectangular drainage area for use with the method of Economides' *et al*¹³ has been extended to account for short horizontal wells ($L/x_e < 0.25$), and high or low aspect ratio drainage shapes ($x_e/y_e > 4$ or $x_e/y_e < 0.25$).

Nomenclature

a	=	flow area, ft ²
A	=	drainage area, acres
B_o	=	formation volume factor, bbl/STB
C_d	=	drag coefficient, dimensionless
C_H	=	shape factor, dimensionless
d_{core}	=	core plug diameter, in
D_p	=	particle diameter, in
F_d	=	drag force, lbf
h	=	net formation thickness, ft
J	=	well productivity, STB/day/psi
J_o	=	undamaged well productivity, STB/day/psi
k	=	effective permeability, md
k_i	=	effective permeability in the i direction, md
L'	=	transformed completion length, ft
P_D	=	dimensionless pressure
P_{wf}	=	flowing bottomhole pressure, psia
Q	=	volumetric flowrate, bbls/day
r_e	=	drainage radius, ft
r_w	=	wellbore radius, ft
S	=	mobile phase saturation, dimensionless
$x_{e'}$	=	transformed drainage length parallel to the completion, ft
$y_{e'}$	=	transformed drainage length perpendicular to the completion, ft
χ	=	mean

- γ = specific gravity, water = 1.00 @ 60 F
 ϕ = porosity, fraction
 μ = viscosity, cp
 ρ = density, lbm/ft³
 σ = standard deviation
 ΣS = skin, dimensionless
 v = velocity, ft/s

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Table 1 - Extended Shape Factor (C_H) Table

		L'/x_e'							
		0.05	0.10	0.25	0.50	0.75	0.90	0.95	1.00
x_e'/y_e'	0.05	26.70	25.24	23.41	22.13	21.39	20.99	20.97	20.95
	0.10	16.20	14.80	12.98	11.65	10.92	10.58	10.50	10.47
	0.20	10.97	9.57	7.74	6.41	5.69	5.37	5.29	5.25
	0.50	7.83	6.42	4.59	3.26	2.53	2.24	2.16	2.09
	1.00	6.79	5.39	3.55	2.21	1.49	1.20	1.12	1.04
	2.00	6.44	5.04	3.19	1.80	1.02	0.70	0.61	0.52
	5.00	7.74	6.31	4.27	2.39	1.13	0.52	0.36	0.25
	10.00	11.55	9.98	7.30	4.14	1.81	0.69	0.39	0.14
	20.00	20.44	18.45	13.96	7.96	3.37	1.21	0.61	0.09

Table 2 – Example Drag Force Results

Well	A	B	B1	B2
Fluid	Oil	Oil	Heavier Oil	Lighter Oil
Description	Lab Test	Lab Test	Hypothetical	Hypothetical
ϕ	22.7%	25.0%	25.0%	25.0%
S_{mobile}	100.0%	100.0%	100.0%	100.0%
d_{core} , in	1.00	1.00	1.00	1.00
D_p , in	7.9E-04	7.9E-04	7.9E-04	7.9E-04
ρ , lbm/ft ³	53.0	53.0	53.0	53.0
Q , bbls/day	6.1E-02	3.7E-02	3.7E-03	3.7E-01
μ_o , cp	4.1	4.1	41.0	0.41
a , ft ²	5.5E-03	5.5E-03	5.5E-03	5.5E-03
v , ft/s	3.2E-03	1.8E-03	1.8E-04	1.8E-02
Re	4.1E-03	2.2E-03	2.2E-05	2.2E-01
C_D	5.9E+03	1.1E+04	1.1E+06	1.1E+02
F_d , lbf	1.7E-10	9.4E-11	9.4E-11	9.4E-11
Critical Rate Exceeded?	No	Yes		

Table 3 - Uncertainty Ranges for Example Calculations (see Table 5)

		Variable	Distribution Type	Min	χ	σ	Max	Shift*
Skin		$k_x=k_y$, md	Log Normal		500	1.5		
		k_z/k_x	Uniform	0.2			0.4	
		μ_o , cp	Normal		.61 or 10.5	10% of χ		
		B_{oi} , bbl/STB	Normal		1.45 or 1.15	2% of χ		
		h , ft	Normal		100	15		
		CHP	Log Normal		7.838	0.635		6
		CHGP	Log Normal		19.202	0.897		6
		CHFP	Log Normal		8.98	0.704		6
		OHGP	Log Normal		14.182	0.47		6

* To facilitate using negative values for skin with a log-normal distribution, the skin data were all shifted six units in the positive direction before developing the distribution, then they are shifted back six units in the negative direction after the distribution is used to generate skin at any particular probability.

Table 4 – Operating Constraints and Impact on Production Issues

		Constraint-Related Issues																	
		Regulatory Requirements	Processing Considerations	Inorganic Precipitation	Microbial Activity	Organic Precipitation	Gas Slugging	Pressure & Fluid Control	Erosion (Other than Screen)	Corrosion	Collapse	Downhole Pump or Motor Failure	Screen Erosion	Gravel or Screen Plugging	Fines Migration	Sand Production (No Screens)	Compaction	Clay Swelling	Injection Out of Zone
Issue Occurrence Region	Surface	♦	♦	♦	♦	♦	♦	♦	♦	♦									
	Subsea Tree / Flowline			♦	♦	♦	♦	♦	♦	♦	♦								
	Upper Completion			♦	♦	♦	♦	♦	♦	♦	♦	♦							
	Lower Completion			♦	♦	♦		♦	♦	♦	♦		♦	♦	♦	♦	♦		
	Formation			♦	♦	♦										♦		♦	♦
Surveillance Type	Constraint Type	Constraint Impact (D=Direct, I=Indirect, N=None or N/A)																	
Surface Measurement	Max. Total Q	N	D	I	N	I	N	N	D	N	I	N	I	I	I	I	N	N	I
	Min. Total Q	N	N	N	N	I	N	N	N	N	N	I	N	N	N	N	N	N	N
	Min. Gas Q	N	D	N	N	N	D	N	N	N	N	N	N	N	N	N	N	N	N
	Max. Temp	N	D	D	N	N	N	I	N	N	N	I	N	N	N	N	N	N	N
	Max. Shut In Pressure	N	D	N	N	N	N	D	N	N	N	N	N	N	N	N	N	N	N
	Min. Temp	N	I	D	N	D	N	N	N	N	N	N	N	I	N	N	N	N	I
	Produced Sand Concentration	D	I	N	N	N	N	I	D	I	N	D	D	D	N	D	N	N	N
	Produced Sand Size Distribution	N	I	N	N	N	N	I	D	I	N	D	D	D	N	D	N	N	N
	Max. GOR	D	D	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
	Max. GLR	N	N	N	N	N	N	N	N	N	N	D	N	N	N	N	N	N	N
	SO ₄ Content	N	D	D	D	N	N	N	N	I	N	I	N	I	N	N	N	N	N
	O ₂ Content	N	D	N	D	N	N	I	N	D	N	N	N	N	N	N	N	N	N
	pH	N	I	D	I	N	N	I	N	D	N	N	N	N	N	N	N	N	I
Salinity	N	D	D	N	N	N	N	N	N	N	N	N	N	N	N	N	N	D	N
Downhole Gauges	Min. Pwf	N	N	D	N	N	I	N	I	N	D	I	I	N	I	D	D	N	N
	Max. Pwf (Injectors)	D	N	D	N	N	N	D	N	N	N	N	N	N	N	N	N	N	D
	Min. Temperature (Injectors)	N	N	D	N	N	N	N	N	N	N	N	N	N	N	N	N	N	D
	Max. Temperature	N	N	N	N	N	N	I	N	N	N	D	N	N	N	N	N	N	N
	DP _{Completion}	N	N	I	N	N	N	N	N	N	N	D	N	I	N	I	D	N	N
	DP _{Formation}	N	N	I	N	N	N	N	N	N	N	N	N	I	N	I	N	D	N
	DP _{Total}	N	N	I	N	N	N	N	N	N	N	N	N	I	N	I	I	I	N
Min. Flow	N	N	I	N	I	N	N	N	N	N	D	N	N	N	N	N	N	N	
Distributed Sensing or Production Logging	Max. Specific Q	N	N	N	N	N	N	N	N	N	N	N	I	I	I	I	N	N	N
	Max. Screen Flux	N	N	N	N	N	N	N	N	N	N	N	D	D	I	N	N	N	N
	Max. Formation Flux	N	N	N	N	N	N	N	N	N	N	N	I	I	I	I	N	N	N
	Drag Force	N	N	N	N	N	N	N	N	N	N	N	N	D	D	I	N	N	N

Table 5 - Calculation Examples and Results

GENERAL DATA

h	100	ft	Screen Open Area	10%	D _p	7.9E-04	in
Base Pipe Outside Diam.	5.500	in	Flowing BHP	2,400	S	75%	
Fluid Type	Oil		Flowing BHT	110	φ	30%	
Reservoir Temp.	120	F	Separator Temp.	80	k _x = k _y	500	md
Reservoir Press.	3,600	psia	Separator Press.	115.0	k _z / k _x	0.30	
Solution GOR	500	SCF/STB	A	640	F _d Limit	5.00E-11	lbf
r _e	2,979	ft	x _e = y _e	5,280	υ Screen Limit	0.25	ft/sec
r _w	0.35	ft					

CASE-SPECIFIC DATA

Case	A	B	C	D	E	F	G	H	I
Well Configuration	Vertical	Vertical	Vertical	Vertical	Vertical	Horizontal	Horizontal	Horizontal	Horizontal
Completion Type	CHP	CHGP	OHGP	CHFP	CHFP	OHGP	OHGP	OHGP	OHGP
API Oil Gravity	40.0	40.0	40.0	40.0	17.0	40.0	17.0	17.0	17.0
R _{si} , SCF/STB	900	900	900	900	300	900	300	300	300
Minimum Local Skin	(1.0)	(1.0)	(1.0)	(4.0)	(4.0)	(1.0)	(1.0)	(1.0)	(1.0)
Total Completion Length, ft	100	100	100	100	100	500	500	5,000	5,000
Effective Completion Length, ft	100	100	100	100	100	500	500	5,000	1,000
Perf. Density, shots/ft	12	12		12	12				
Formation Penetration, in	16.0	6.0		6.0	6.0				
Entrance Hole Diameter, in	0.35	0.70		0.70	0.70				
Frac Half-Length, ft				50	50				
Frac Height, ft				100	100				
P _{sat} , psia	2,911	2,911	2,911	2,911	2,498	2,911	2,498	2,498	2,498
μ _o (reservoir), cp	0.61	0.61	0.61	0.61	10.51	0.61	10.51	10.51	10.51
Deterministic J at Min Local Skin, bbls/day/psi	55.0	55.0	55.0	93.5	6.8	107.4	7.9	45.4	45.4
J Factor Partial Completion	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.74
Corrected J Partial Completion	55.0	55.0	55.0	93.5	6.8	107.4	7.9	45.4	33.6
Flow Convergence Factor	1.0	1.0	1.0	1.0	1.0	4.0	4.0	1.0	3.7
Sand Face Area Basis	Minimum Area: Hypothetical Enlarged OH or Sum of Conical Perf Areas	Minimum Area: Hypothetical Enlarged OH or Hemi-spherical Cavities	Open Hole	Elliptical Bi-Wing Frac, 2 Faces	Elliptical Bi-Wing Frac, 2 Faces	Open Hole	Open Hole	Open Hole	Open Hole
Sand Face Flux Area, ft ²	7.331E+01	5.4E+02	2.2E+02	1.6E+04	1.6E+04	1.1E+03	1.1E+03	1.1E+04	1.1E+04
Relative Sand Face Flux Area	0.0047	0.0342	0.0142	1.0000	1.0000	0.0708	0.0708	0.7083	0.7083
Screen Open Flow Area, ft ²		14.4	14.4	14.4	14.4	72.0	72.0	719.9	144.0
Relative Screen Open Area		0.020	0.020	0.020	0.020	0.100	0.100	1.000	0.200
Collapse Limit, psi	4,000	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500
Collapse Basis	Perf. Casing	Screen	Screen	Screen	Screen	Screen	Screen	Screen	Screen
Min P _{wf} (Collapse), psia	-	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100
Min P _{wf} (Fines Migration), psia	3,571	3,382	3,512	-	-	3,544	3,556	3,275	3,484
Min P _{wf} (Screen Erosion), psia	-	2,576	2,576	3,008	-	-	-	-	-
Min P _{wfr} , psia	3,571	3,382	3,512	3,008	1,100	3,544	3,556	3,275	3,484
Max Total Drawdown, psi	29	218	88	592	2,500	56	44	325	116
Limiting Factor	Fines	Fines	Fines	Screen Erosion	Collapse	Fines	Fines	Fines	Fines

Probabilistic Productivities

p10 J, bbls/d/psi	16.7	5.3	11.2	13.2	1.0	19.7	1.2	9.7	7.2
p50 J, bbls/d/psi	28.7	17.9	23.8	34.5	2.5	43.0	2.7	20.6	15.2
p90 J, bbls/d/psi	86.6	53.5	49.7	79.1	5.9	94.2	6.0	44.2	32.7

Maximum Rates Based on Combined Pwf Constraint & Productivity Forecast

p10 Q @ Max Pwf, bbls/day	484	1,155	986	7,814	2,152	1,103	53	3,153	833
p50 Q @ Max Pwf, bbls/day	832	3,902	2,094	20,424	5,381	2,408	119	6,695	1,768
p90 Q @ Max Pwf, bbls/day	2,511	11,663	4,374	46,827	12,699	5,275	264	14,365	3,794

Appendix A - Skin Data

ID	Status	Completion	Detail	Shunts?	Source	Skin	Perm, mD	ID	Status	Completion	Detail	Shunts?	Source	Skin	Perm, mD
1	P	CHGP	Unknown	N	SPE 30470	247.00	498.0	85	P	CHFP	Unknown	N	SPE 30470	4.70	
2	P	CHGP	Unknown	N	SPE 30093	100.00	8200.0	86	P	CHFP	Unknown	N	SPE 30470	4.58	29.5
3	P	CHGP	Unknown	N	SPE 30470	96.99	130.6	87	P	CHFP	Unknown	N	SPE 30470	4.54	11.5
4	P	CHGP	Unknown	N	SPE 30470	62.00	5400.0	88	P	CHGP	Unknown	N	SPE 30470	4.49	1756.1
5	P	CHGP	Unknown	N	SPE 30470	61.35	3356.1	89	P	CHGP	Unknown	N	SPE 30470	4.41	220.4
6	P	CHGP	Unknown	N	SPE 30470	57.91	191.2	90	P	CHFP	XL Gel Frac	N	SPE 73722	4.20	
7	P	CHFP	Unknown	N	SPE 36423	56.00	850.0	91	P	CHFP	Unknown	N	SPE 30470	4.00	10.2
8	P	CHFP	Unknown	N	SPE 36423	56.00	1500.0	92	P	CHF	XL Oil Frac	N	SPE 22750	4.00	76.0
9	P	CHGP	Unknown	N	SPE 30093	54.00	3600.0	93	P	CHFP	XL Gel Frac	N	SPE 73722	3.90	
10	I	CHGP	VES	Y	SPE 77434	50.00		94	P	CHFP	XL Gel Frac	N	SPE 73722	3.90	
11	I	CHGP	VES	Y	SPE 77434	50.00		95	P	CHGP	Unknown	N	SPE 30470	3.89	90.6
12	P	CHGP	Unknown	N	SPE 30093	50.00	1010.0	96	P	CHFP	XL Gel Frac	N	SPE 73722	3.80	
13	P	CHFP	Unknown	N	SPE 36423	47.00	189.0	97	P	CHGP	HRWP	N	SPE 38592	3.54	
14	P	CHFP	XL Gel Frac	N	SPE 73722	45.20		98	P	CHFP	Unknown	N	SPE 30470	3.36	73.7
15	P	CHGP	Unknown	N	SPE 30470	41.31	119.1	99	P	CHFP	Unknown	N	SPE 36423	3.30	855.0
16	P	CHGP	Unknown	N	SPE 30470	41.31	119.1	100	P	CHGP	Unknown	N	SPE 30470	3.21	24.0
17	P	CHGP	VES	Y	SPE 77434	40.00		101	P	CHFP	XL Gel Frac	N	SPE 73722	3.10	
18	P	CHGP	Unknown	N	SPE 30093	38.00	9603.0	102	P	CHFP	Linear Gel	N	SPE 30115	3.10	161.5
19	P	CHP	Unknown	N	SPE 30470	37.24	2.4	103	P	CHFP	Water Frac	N	SPE 36459	3.00	349.0
20	P	CHGP	VES	Y	SPE 77434	36.00		104	P	CHFP	XL Gel Frac	N	SPE 73722	2.60	
21	P	CHGP	Unknown	N	SPE 30093	36.00	860.0	105	P	CHFP	Unknown	N	SPE 30470	2.59	13.2
22	P	CHFP	Unknown	N	SPE 36423	34.00	1500.0	106	P	CHGP	Unknown	N	SPE 30470	2.41	5.0
23	P	CHGP	Unknown	N	SPE 30093	33.70	6794.0	107	P	CHFP	XL Gel Frac	N	SPE 73722	2.30	
24	P	CHGP	HRWP	N	SPE 38592	32.64		108	P	CHFP	Unknown	N	SPE 30470	2.01	48.2
25	P	CHGP	Unknown	N	SPE 30470	31.80	15.9	109	P	CHFP	XL Gel Frac	N	SPE 73722	1.90	
26	P	CHGP	Unknown	N	SPE 30470	31.58	137.2	110	P	CHFP	XL Gel Frac	N	SPE 73722	1.80	
27	P	CHGP	VES	Y	SPE 77434	31.00		111	P	CHGP	Unknown	N	SPE 30470	1.73	16.6
28	P	CHFP	Unknown	N	SPE 36423	31.00	1688.0	112	P	CHFP	XL Gel Frac	N	SPE 73722	1.60	
29	P	CHGP	Unknown	N	SPE 30470	30.50	142.0	113	P	CHFP	Unknown	N	SPE 30470	1.60	62.3
30	P	CHGP	Unknown	N	SPE 30470	29.66	91.5	114	P	CHFP	XL Gel Frac	N	SPE 38592	1.58	
31	P	CHGP	Unknown	N	SPE 30470	28.91		115	P	CHFP	XL Gel Frac	N	SPE 73722	1.50	
32	P	CHFP	Unknown	N	SPE 30470	27.12	13.7	116	P	CHFP	XL Gel Frac	N	SPE 73722	1.40	
33	P	CHFP	XL Gel Frac	N	SPE 38592	26.10		117	P	CHFP	Unknown	N	SPE 30470	1.32	
34	P	CHFP	XL Gel Frac	N	SPE 73722	26.00		118	P	CHFP	Linear Gel	N	SPE 30115	1.30	141.7
35	P	CHFP	Linear Gel	N	SPE 36459	25.00	1206.0	119	P	CHFP	XL Gel Frac	N	SPE 73722	1.20	
36	P	CHGP	HRWP	N	SPE 38592	23.10		120	P	CHFP	Linear Gel	N	SPE 30115	1.20	363.2
37	P	CHGP	Unknown	N	SPE 30093	23.00	1150.0	121	P	OHGP	Unknown	N	SPE 58731	1.10	4100.0
38	P	CHFP	XL Gel Frac	N	SPE 73722	21.90		122	P	CHGP	Unknown	N	SPE 30470	1.05	51.2
39	P	CHFP	Unknown	N	SPE 30470	21.89	48.4	123	P	CHFP	XL Gel Frac	N	SPE 73722	1.00	
40	P	CHGP	Unknown	N	SPE 30470	20.84		124	P	CHFP	Water Frac	N	SPE 36459	1.00	57.0
41	P	CHGP	Unknown	N	SPE 30093	20.00	2160.0	125	P	CHGP	Unknown	N	SPE 30093	1.00	215.0
42	P	CHGP	Unknown	N	SPE 30093	20.00	8490.0	126	P	CHFP	XL Gel Frac	N	SPE 73722	0.80	
43	P	CHGP	Unknown	N	SPE 30093	19.80	5136.0	127	P	CHFP	XL Gel Frac	N	SPE 38592	0.75	
44	P	CHFP	XL Gel Frac	N	SPE 73722	19.40		128	P	CHP	Unknown	N	SPE 30470	0.65	1.6
45	P	CHFP	XL Gel Frac	N	SPE 73722	18.00		129	P	CHFP	Unknown	N	SPE 30470	0.59	124.5
46	P	CHGP	Unknown	N	SPE 30093	16.10	6180.0	130	P	CHFP	XL Gel Frac	N	SPE 38592	0.40	
47	P	CHFP	Linear Gel	N	SPE 36459	16.00	1312.0	131	P	CHFP	Unknown	N	SPE 30470	0.14	0.9
48	P	CHFP	XL Gel Frac	N	SPE 73722	15.90		132	P	CHFP	Unknown	N	SPE 30470	0.14	297.5
49	P	CHFP	Linear Gel	N	SPE 36459	15.00	1098.0	133	P	OHGP	MOBM DIF	Y	SPE 77434	0.00	
50	P	CHFP	Linear Gel	N	SPE 30115	15.00	154.6	134	P	CHFP	XL Gel Frac	N	SPE 73722	0.00	
51	P	CHGP	Unknown	N	SPE 30093	14.70	4068.0	135	P	CHFP	XL Gel Frac	N	SPE 73722	0.00	
52	P	CHP	Unknown	N	SPE 30470	14.45	56.6	136	P	CHFP	XL Gel Frac	N	SPE 73722	0.00	
53	P	CHFP	XL Gel Frac	N	SPE 73722	13.90		137	P	CHFP	XL Gel Frac	N	SPE 73722	0.00	
54	P	CHGP	Unknown	N	SPE 30470	13.60	925.1	138	P	CHFP	XL Gel Frac	N	SPE 73722	0.00	
55	P	CHGP	VES	Y	SPE 77434	13.00		139	P	CHFP	Unknown	N	SPE 30470	-0.03	0.3
56	P	CHFP	Unknown	N	SPE 30470	12.85	251.6	140	P	CHFP	Unknown	N	SPE 30470	-0.20	169.0
57	P	CHGP	Unknown	N	SPE 30093	12.00	2643.0	141	P	OHGP	MOBM DIF	Y	SPE 77434	-0.40	
58	P	CHGP	Unknown	N	SPE 30470	11.88	94.3	142	P	CHGP	Unknown	N	SPE 30470	-0.56	8.9
59	P	CHFP	Unknown	N	SPE 36423	11.30	833.0	143	P	OHGP	MOBM DIF	Y	SPE 77434	-0.70	
60	P	CHGP	HRWP	N	SPE 38592	11.00		144	P	CHFP	XL Gel Frac	N	SPE 73722	-0.70	

ID	Status	Completion	Detail	Shunts?	Source	Skin	Perm, mD		ID	Status	Completion	Detail	Shunts?	Source	Skin	Perm, mD
61	P	CHFP	XL Gel Frac	N	SPE 73722	10.20			145	P	CHP	Unknown	N	SPE 30470	-0.75	238.5
62	P	CHFP	Unknown	N	SPE 36423	9.70	904.0		146	P	CHP	Unknown	N	SPE 30470	-0.76	66.0
63	P	CHFP	Unknown	N	SPE 36423	8.70	895.0		147	P	CHFP	XL Gel Frac	N	SPE 77434	-0.80	
64	P	CHGP	Unknown	N	SPE 30470	8.69	111.6		148	P	CHFP	XL Gel Frac	N	SPE 73722	-1.00	
65	P	CHFP	XL Gel Frac	N	SPE 73722	8.20			149	P	CHFP	Unknown	N	SPE 30470	-1.10	
66	P	CHGP	Unknown	N	SPE 30093	8.20	3039.0		150	P	CHFP	Unknown	N	SPE 30470	-1.20	4.3
67	P	CHFP	Unknown	N	SPE 36423	8.10	1500.0		151	P	CHFP	Linear Gel	N	SPE 30115	-1.40	169.6
68	P	CHGP	Unknown	N	SPE 30093	7.80	4673.0		152	P	CHFP	XL Gel Frac	N	SPE 73722	-1.50	
69	P	CHGP	Unknown	N	SPE 30470	7.74	3596.3		153	P	CHF	XL Oil Frac	N	SPE 22750	-1.50	7.5
70	P	CHGP	Unknown	N	SPE 30470	7.70	1221.0		154	P	CHP	Unknown	N	SPE 30470	-1.52	51.8
71	P	CHGP	Slurry Pack	N	SPE 36423	7.30	1374.0		155	P	CHFP	XL Gel Frac	N	SPE 73722	-1.70	
72	P	CHFP	Linear Gel	N	SPE 36459	7.00	156.0		156	P	CHGP	Unknown	N	SPE 30470	-1.87	7.3
73	P	CHFP	Water Frac	N	SPE 36459	7.00	430.0		157	P	CHFP	Linear Gel	N	SPE 30115	-1.90	286.2
74	P	CHGP	Unknown	N	SPE 30470	6.90	83.0		158	I	CHFP	XL Gel Frac	N	SPE 77434	-2.00	
75	P	CHGP	Unknown	N	SPE 30470	6.30	519.9		159	P	CHGP	HRWP	N	SPE 38592	-2.13	
76	P	CHFP	XL Gel Frac	N	SPE 73722	6.00			160	P	CHFP	Unknown	N	SPE 30470	-2.17	40.2
77	P	CHGP	Unknown	N	SPE 30093	5.90	1170.0		161	P	CHFP	Unknown	N	SPE 30470	-2.42	2.2
78	P	CHFP	Unknown	N	SPE 30470	5.51	218.2		162	P	CHFP	XL Gel Frac	N	SPE 73722	-2.60	
79	P	CHFP	Linear Gel	N	SPE 36423	5.20	1181.0		163	P	CHFP	Unknown	N	SPE 30470	-2.69	
80	P	CHFP	Unknown	N	SPE 30470	5.16	142.8		164	P	CHFP	Linear Gel	N	SPE 30115	-2.70	283.8
81	P	OHGP	MOBM DIF	Y	SPE 77434	5.00			165	P	CHFP	Unknown	N	SPE 30470	-2.71	0.2
82	P	CHFP	Water Frac	N	SPE 36459	5.00	18.0		166	P	CHF	XL Oil Frac	N	SPE 22750	-2.90	161.0
83	P	CHF	XL Oil Frac	N	SPE 22750	5.00	98.0		167	P	CHFP	Unknown	N	SPE 30470	-3.41	17.7
84	P	CHFP	XL Gel Frac	N	SPE 73722	4.80			168	P	CHFP	VES	N	SPE 38592	-3.80	

Abbreviations

Status

P - Producer
I - Injector

Completion

CHF - Cased Hole Fractured
CHFP - Cased Hole Frac Pack
CHGP - Cased Hole Gravel Pack
CHP - Cased Hole Perforated
OHGP - Open Hole Gravel Pack

Detail

MOBM DIF - Mineral Oil-Based Mud Drill-In Fluid
HRWP - High-Rate Water Pack
VES - Visco-Elastic Surfactant
XL - Cross-Linked